

App Dev 1 Project Report

1. Student Details

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About Me : I am a student of the IIT Madras BS Degree program with an interest in programming and web development. I enjoy learning new technologies and building simple applications to solve real-world problems.

2. Project Details

Project Title : Trekking Management Application

Problem Statement:

To design and develop a web-based Trekking Management Application that helps manage treks, users, staff, and bookings in one place. The system allows users to browse and book treks, while admins and staff can manage treks, bookings, participants, and trek activities efficiently.

Approach:

The application was developed using Flask as the backend framework with SQLite as the database. It provides separate features for Admin, Staff, and Trekker users. Users can register, log in, view available treks, and make bookings, while Admin and Staff can manage treks, bookings, participants, and trek details.

3. AI/LLM Declaration

I used ChatGPT to assist with understanding concepts, debugging errors, improving code, and designing some parts of the project. The extent of AI/LLM usage is around **30-35%**. The AI assistance was used as a learning and development aid. I reviewed and integrated the code into my project and tested the implemented features.

4. Technologies and Frameworks Used

Technology / Library	Purpose
Python	Main programming language
Flask	Backend web framework
SQLAlchemy	Database management and ORM
SQLite	Storing users, treks, bookings and other project data
HTML	Creating the web pages
CSS	Styling and designing the web pages
Jinja2	Displaying dynamic data in HTML pages

5. Database Schema / ER Diagram

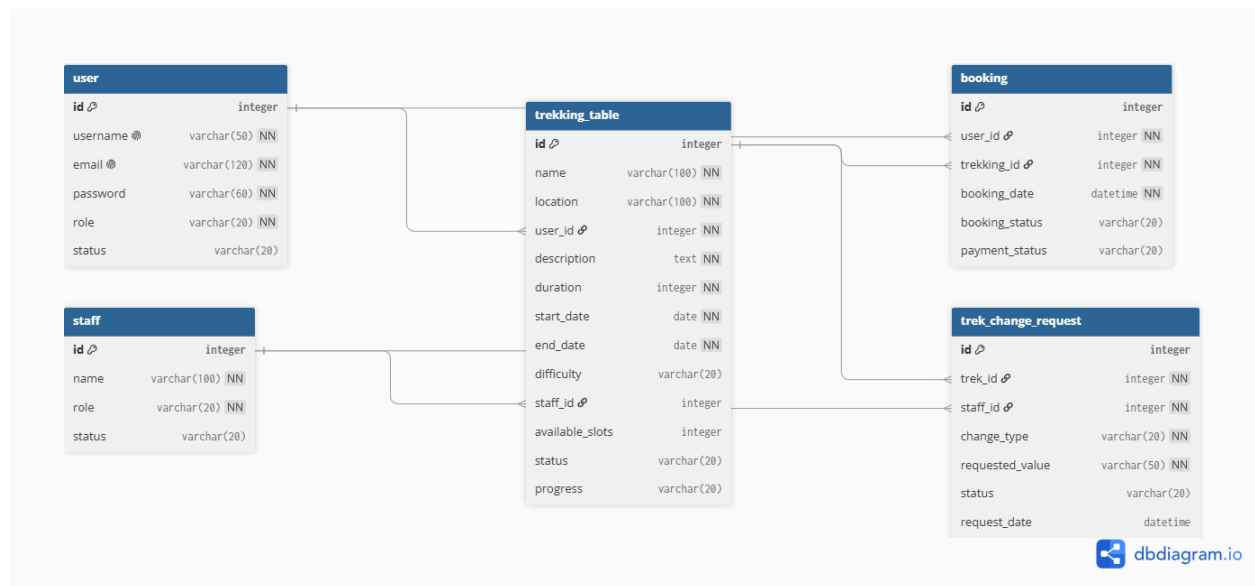
Tables

1. **User** — stores user details such as username, email, password, role, and status.
2. **Trekking_table** — stores trek details such as trek name, location, description, duration, available slots, assigned staff, status, and progress.
3. **Booking** — stores booking details such as user, trek, booking date, and booking status.
4. **Staff** — stores staff details such as name, role, and status.

Relationships

- **User → Booking: One user can make multiple bookings.**
- **Trekking_table → Booking: One trek can have multiple bookings.**
- **Staff → Trekking_table: One staff member can be assigned to multiple treks.**

E-R Diagram :



6. Application Endpoints

Endpoint	Method	Description
/	GET	Displays the home page
/signup	GET, POST	Allows a new user to create an account
/login	GET, POST	Allows users to log in
/logout	GET	Logs the current user out
/browse_treks	GET	Displays available treks
/book_trek/<id>	GET, POST	Allows a user to book a trek
/my_bookings	GET	Displays the user's bookings
/admin_dashboard	GET	Displays the admin dashboard and project statistics

/create_trek	GET, POST	Allows the admin to add a new trek
/assign_trek	GET, POST	Assigns staff to a trek
/staff_dashboard	GET	Displays the staff dashboard
/participants	GET	Displays participants of assigned treks

7. Video Presentation

Drive Link:

 Screen Recording 2026-08-09 113244.mp4